

MIAE 221 · Rapid Review Sheet · Part 5B

Grain Boundaries, Stacking Faults, Volume Defects & Microscopy (Lecture 8)

1. Polycrystals & Grain Boundaries (2-D defects)

- **Solidification:** random nuclei → grow → impinge → **grains** separated by **grain boundaries**. Most engineering materials are **polycrystalline**.
- **High-angle** GB: large misorientation. **Low-angle** GB: a few degrees, = **array of dislocations** — **tilt** (edge), **twist** (screw); $\theta \approx b/D$.
- GB atoms: less bonded, lower packing, lower coordination → **grain boundary energy** (J/m²) → GBs are **more reactive**, attract **impurity segregation**, are fast diffusion paths, and **grow** on heating.
- **Fine grains** = more GB area = stronger (Hall–Petch). **Coarse grains** = less GB area.

2. Observing Grains & Grain Size

- **Macrostructure:** visible to the naked eye (lead ingot, galvanized zinc). **Microstructure:** needs a microscope. Camera image = **photomicrograph**.
- **Preparation:** grind → polish (mirror, featureless) → **etch** (chemical reagents attack GBs preferentially → grooves **scatter light** → dark lines = **optical contrast**).
- **ASTM grain size** (at 100×): $N = 2^{G-1}$; other magnification: $N_M (M/100)^2 = 2^{G-1}$. **Larger G = smaller grains**.
- **Intercept method:** $\bar{\ell} = \frac{L_T}{P M}$; $G = -6.6457 \log_{10} \bar{\ell} - 3.298$ ($\bar{\ell}$ in mm).

3. Stacking Faults (planar)

- FCC: ...**ABCABC**...; HCP: ...**ABAB**.... Fault = wrong plane order, e.g. ...**ABCAB|A**... (C missing) → thin **HCP-like** layer in FCC.
- Atoms stay in close-packed sites (12 neighbours) → **low energy** → **common**. Form during **solidification** and **plastic deformation**.
- **Intrinsic** (plane removed) vs. **extrinsic** (plane inserted). **Low SFE** (stainless, brass) → wide partials, little cross-slip, high work hardening; **high SFE** (Al) → easy cross-slip.

4. Volumetric (3-D) Defects

Defect	Origin	Effect
Inclusions	Foreign particles (oxides, slag)	Stress raisers, lower toughness
Precipitates	Second phase from heat treatment	Usually strengthen
Pores / voids	Gas, shrinkage, sintering	Lower strength and density
Cracks	Thermal/mechanical stress	Most dangerous; fracture initiation

5. Microscopy

- Resolution limited by wavelength: light $\lambda \approx 500$ nm (max $\approx 2000\times$); electrons $\lambda \approx 0.003$ nm (up to $\approx 1\,000\,000\times$).

Microscope	Resolution	Key points
Human eye	≈ 0.1 mm ($\sim 10^6$ Å)	Macrostructure
OLM	3000 Å	Reflected light; polished + etched; shallow depth of field
SEM	10–50 Å	Scanned beam; surfaces, fractography; large depth of field ; EDS composition
TEM	2–5 Å , near atomic	Electrons through thin (electron-transparent) samples; dislocations, GBs, diffraction contrast
SPM (STM/AFM)	≈ 1 Å, atomic	Sharp probe + piezo scanner + feedback; computer-built 3-D topography

- **Why SPM only in the 1980s:** TEM (≈ 1938) and SEM (≈ 1965) were **analog** (photographic film); SPM **depends on computers** to build the image.
- **Clicker answers:** etched grains visible → **C** (GBs scatter light); rough fracture surface in focus in SEM → **A** (depth of field).